Anjali Herbal Life Ayurveda India LLP — Final Application Report

Date: 15 Oct 2026 · Environment inspected: PREVIEW (DB anjali_ayurveda). Production (anjaliherbal.life) runs its own database and was NOT read or modified.

A. Executive summary

- Financial engine corrected to the authoritative rule: Company 30% of S, Commission Pool 70% of S, GD–MDD positional rates applied to the pool (45% of pool paid, 55% of pool retained — not normalised). Mandated ₹1,500 test passes exactly (BD ₹157.50 … total ₹472.50, company ₹1,027.50).
- Team Purchase Income now pays 2.5% of S to every eligible GREEN upline in the first nine actual positions (was level-1 only).
- New auditable refund engine: full/partial refunds, proportional clawback, paid-cancellation only reverses when a payment return is recorded, idempotent per refund reference, over-refund blocked, activation revert on full refund, BV reduces with refunds.
- 60/60 isolated financial tests PASS (backend/tests/test_financial_engine_isolated.py, runs on a throwaway DB, drops it afterwards).
- Cleanup: 2 user IDs removed (confirmed synthetic fixtures) + orphan residue of 2 previously deleted test IDs. All genuine users preserved. No collection was truncated or reset.
- Overall status: WORKING in preview; production needs redeploy + two secrets (see K).

B. Inventory

Backend (FastAPI, /app/backend): server.py (bootstrap, catalogue seed only when catalogue empty, admin seed from env, demo seeder permanently removed), core.py (db, wallet ledger add_wallet_transaction, audit, settings, referral tree), auth_routes.py (OTP/password/Google/admin login), shop_routes.py (products, orders, Razorpay/Stripe/UPI/manual payments, _process_paid_order, 70%-pool matrix, repurchase, team purchase, refund engine), wallet_routes.py (wallet, team, KYC, withdrawals, dashboard, rank/BV), admin_routes.py (users, approvals, orders, refunds, KYC, withdrawals, MFI, club income, income rules, reports, audit), income_config.py (versioned rate configs), club_income.py (Royalty/Master/Super Club), cron_routes.py (monthly cron), providers.py (Twilio), storage.py (object storage).

Frontend (React, /app/frontend/src/pages): Landing, Login, Register, AuthCallback, Dashboard, Products, ProductDetail, Cart, Checkout, Orders, OrderDetail, Wallet, Withdrawals, Referral, Team (Ranks/Matrix/Tree/List), KYC, Profile, Notifications, Legal; admin: Dashboard, Users, Products, Orders (+Refund), KYC, Withdrawals, Commissions, MonthlyIncome (+Club income), IncomeRules, Reports, Settings, AuditLogs, SecureLogin.

Database collections: users, registration_requests, referral_relationships, referral_codes, orders, order_refunds (new), payment_transactions, commissions, wallets, wallet_transactions, withdrawal_requests, kyc_records, activation_records, notifications, audit_logs, business_settings, income_rule_configs, monthly_income_runs, cron_runs, products, product_categories, counters, otp_*, files.

Integrations: Razorpay (TEST keys), Twilio SMS (trial-blocked), Emergent Google Sign-In, Emergent Object Storage, Postal pincode API, platform cron (.emergent/crons.yml).

C. Feature status

Feature	Status
Registration → admin approval → Distributor ID	WORKING
₹100 direct referral at approval (once)	WORKING (test T8)
Activation ≥₹1,500 admin-confirmed / Razorpay	WORKING (T1, T6)
45%-of-70%-pool positional matrix	WORKING (T1, T2)
Personal Repurchase 5% (skip first)	WORKING (T3)
Team Purchase 2.5% × 9 positions	WORKING (T1, T2)
Monthly Fixed Income 1% equal split	WORKING (existing, T9)
Royalty/Master/Super Club	WORKING per your 13-Oct decision (own-team turnover, exact rank, monthly cron)
Refund full/partial/paid-cancel	WORKING (T4–T7)
Wallet ledger/reconciliation	WORKING (all IDs ledger == wallet)
Withdrawals (GREEN + verified KYC + atomic debit)	WORKING — KEEP, no change
Team/Matrix/Tree screens	WORKING (real downline, IDs + names)
Twilio OTP	BLOCKED (trial account)
Razorpay live	PENDING (test keys)
Legal page content	NOT IMPLEMENTED (placeholder text)
Production data verification	NOT VERIFIED (no access from preview)

D. Financial engine (new)

S = subtotal excl. GST. company_direct = 30%·S, pool = 70%·S, position k gets rate_k · pool if that actual sponsor ancestor is GREEN, else ₹0 (retained). Stored per order in orders.allocation {subtotal, company_direct, commission_pool, rank_commissions_paid, pool_retained, company_total}. Rounding: Decimal ROUND_HALF_UP to paise (money2).

Mandated example (S ₹1,500, GST ₹75, total ₹1,575): company 450 · pool 1,050 · GD 0 · BD 157.50 · AD 52.50 · CD 52.50 · PD 52.50 · MD 52.50 · RD 42 · DD 31.50 · MDD 31.50 · commissions 472.50 · retained 577.50 · company total 1,027.50 · 1,027.50 + 472.50 = 1,500 (test T1).

Other plans unchanged and separate: referral ₹100, repurchase 5% of S, team 2.5% of S per eligible position, MFI 1% pool, clubs per rule.

E. Records removed (preview DB only)

Record	Evidence	Removed
Users ANJ000219 "FRN C", ANJ000220 "FRN D"	Created 2026-09-24 20:32:16 in the same second as ANJ000218 by actor=None during a scripted test burst (15 purchase commissions in 2 s); no registration_requests; sponsor ANJ000218 never existed as a user; no orders/KYC/withdrawals; placeholder names	2 users, 2 wallets, 1 ledger txn (₹100 test referral), 1 commission, 2 relationships, 1 notification
Orphan ₹100 referral credit to ANJ000000 for ANJ000240	ANJ000240 approved+deleted during the 2026-10-11 testing-agent run; user record absent; no relationship	1 commission, 1 ledger txn; ANJ000000 wallet –₹100 (200 → 100)
Orphan wallet bdcf9a00…	Owner user no longer exists, balance ₹0	1 wallet

Total user IDs removed: 2. Cleanup logged in audit_logs id cleanup-2026-10-15. Kept for your review (ambiguous, not deleted): none remaining.

F. Preservation confirmation

Remaining users: ANJ000000, ANJ000215 (Rahul Sharma), ANJ000216 (Priya Verma), ADMIN001 (super admin). Master ID MST000001 referral code intact. Relationships MST000001→ANJ000215→ANJ000216 intact. Order ORD000076 (₹1,500 activation of ANJ000215) and its ledger intact. No delete_many({}), no collection drop, no wallet reset on the preview DB; test scripts only ever touch throwaway DBs (*_tmp) which they drop.

G. Reconciliation (preview DB after cleanup)

ANJ000000 wallet ₹100 = 1 referral credit (ANJ000215) · ANJ000215 wallet ₹100 = 1 referral credit (ANJ000216) · ANJ000216 ₹0 · ADMIN001 ₹0 · 0 duplicate commission groups · 0 withdrawals · 0 orphan records.

H. Code quality & security findings

- FIXED: _process_paid_order non-atomic paid claim (concurrent Razorpay verify + webhook) → now aborts unless it wins the atomic claim (test: 4 concurrent → 1 activation record).
- FIXED: dashboard/team income counted REVERSED commissions.
- FIXED: status=refunded via plain status change reversed everything without a refund reference — now blocked; Refund action requires reference + amount.
- FIXED: float rounding → Decimal half-up for all new money math.
- OK: all admin routes behind require_admin; withdrawals atomic; ledger immutable (originals never deleted; clawbacks linked by reference = order:refund:<ref>).
- NOTE: admin DELETE /orders/{id} hard-deletes an order (pre-existing). Recommend disabling for paid orders — not changed without approval.

I. Changes (files/functions)

- backend/shop_routes.py: COMPANY_DIRECT_PCT, COMMISSION_POOL_PCT, money2, commission_pool; _distribute_purchase_commission (pool base, allocation snapshot); _credit_team_purchase (9 positions); _credit_personal_repurchase (rounding); reverse_order_financials (partial/idempotent/paid-cancel); _process_paid_order (atomic claim).
- backend/admin_routes.py: update_order_status (refund via status blocked, cancel never auto-reverses); new POST /api/admin/orders/{id}/refund (RefundBody).
- backend/wallet_routes.py: _personal_bv (refund-aware), dashboard label, CREDITED-only income.
- frontend/src/pages/admin/AdminOrders.js: Refund button (amount, mandatory reference, optional cancel).
- backend/tests/test_financial_engine_isolated.py: 60-assertion isolated suite.

J. Test evidence

Isolated suite: 60 PASS / 0 FAIL — covers exact ₹1,500 chart, 9-position chain, ineligible positions + company retention, GST exclusion, concurrent payment ×4, activation, first-purchase repurchase exclusion, repurchase on 2nd order, team purchase 9/7 positions, partial refund 50%, duplicate refund reference rejected, remaining refund, over-refund blocked, repeated reversal, paid-cancel without return keeps credits, activation refund → RED, unpaid cancelled → 0 credits, referral idempotency via real approve flow, MFI exclusion, BV before/after refunds, ledger == wallet for every ID.

API checks (preview, admin JWT): refund without reference → 400; status=refunded → 400 guidance; over-amount → 400; dashboard breakdown label updated.

Startup after cleanup: backend healthy (/api/products 200). Reseed prevention: demo seeder removed from code; catalogue seed only when empty; test suite uses throwaway DB.

NOT RUN: real Razorpay webhook (test keys only), real SMS, production DB reconciliation.

K. Remaining work / production limits

1. Redeploy to apply; add production secret WEBHOOK_CRON_SECRET (for the monthly club-income cron).
2. Production data was NOT inspected; any production test IDs need the same evidence-based review there (I can run the same read-only inspection via the deployer diagnostics on request).
3. Twilio upgrade; Razorpay live keys; Legal page content.

L. Decisions needed from you

1. Historical orders in PRODUCTION paid under the old 45%-of-S formula: leave as-is (recommended, no retroactive change) or report affected records for manual adjustment?
2. Disable hard-delete of PAID orders in admin (recommended)?
3. admin DELETE /orders aside, no other approvals pending.